Proceedings of the Human Factors and Ergonomics Society Annual Meeting
http://pro.sagepub.com/

The System Usability Scale: Beyond Standard Usability Testing

Rebecca A. Grier, Aaron Bangor, Philip Kortum and S. Camille Peres

Proceedings of the Human Factors and Ergonomics Society Annual Meeting 2013 57: 187
DOI: 10.1177/1541931213571042

The online version of this article can be found at:
http://pro.sagepub.com/content/57/1/187

Published by:
SAGE
http://www.sagepublications.com

On behalf of:


Human Factors and Ergonomics Society

Additional services and information for *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* can be found at:

Email Alerts: http://pro.sagepub.com/cgi/alerts

Subscriptions: http://pro.sagepub.com/subscriptions

Reprints: http://www.sagepub.com/journalsReprints.nav

Permissions: http://www.sagepub.com/journalsPermissions.nav

Citations: http://pro.sagepub.com/content/57/1/187.refs.html

>> Version of Record - Sep 30, 2013

What is This?

Downloaded from pro.sagepub.com at HFES-Human Factors and Ergonomics Society on June 3, 2014

The System Usability Scale: Beyond Standard Usability Testing

Rebecca A. Grier[1], Aaron Bangor[2], Philip Kortum[3], S. Camille Peres[4]
[1]Institute for Defense Analyses, [2] AT&T Labs, [3] Rice University, [4]Texas A&M University

The system usability scale (SUS; Brooke, 1996) is an instrument commonly utilized in usability testing of commercial products. The goal of this symposium is to discuss the validity of the SUS in usability tests and beyond. This article serves as an introduction to the symposium. Specifically, it provides an overview of the SUS and discusses research questions currently being pursued by the panelists. This current research includes: defining usability norms, assessing usability without performing tasks, and the use of SUS for ergonomics. In addition to this paper, there are four other papers in the symposium, which discuss the impact of experience on SUS data, the relationship between SUS and performance scores, the linkage between SUS and business metrics, as well as the potential for using SUS in test and evaluation for military systems.

INTRODUCTION

Brooke (1996) developed the System Usability Scale (SUS) as a "quick and dirty" measure of usability. The survey consists of ten questions; each has a five point Likert response continuum. The ten questions are as follows:

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

The SUS provides a score from 0 to 100. According to Bangor, Kortum, and Miller (2008) a score of 85 or higher represents exceptional usability and a score below 70 represents unacceptable usability. Several studies have shown the SUS to be highly reliable (Borsci, Federici, & Lauriola, 2009; Bangor et al, 2008). Furthermore, the SUS appears to be valid as a metric of commercial system usability for the tasks performed in the test (Sauro & Lewis, 2012; Kortum, Grier, & Sullivan, 2010; Bangor et al, 2008, 2009). It is our belief that that the SUS may be a useful measure not only in traditional commercial usability tests, but also in other contexts as well. In this symposium, we discuss the evidence that leads us to these hypotheses and the studies that need to be performed to validate those uses. More specifically, recent reports by Sauro and Lewis (2012) suggest that the Bangor et al. (2008) scales are inaccurate. We begin our examination of SUS validity with a discussion of these findings. This is followed by examinations of using the SUS in situations outside of the standard usability test.

USING THE SUS TO ASSIGN A "LETTER GRADE"

One of the main benefits of the SUS is that its output is a seemingly easy-to-understand score, ranging from 0 to 100, with larger scores being better. This unit-less score works very well for making relative comparisons, but as discussed in Bangor et al. (2008), there is often a need to explain what a single SUS score itself really means (i.e., is a 68.5 good or bad?). To address this, Bangor et al. (2008, 2009) presented an adjective-anchored rating scale (e.g., Awful, OK, Excellent, etc.) that correlated highly with SUS scores. Another common scale for evaluation was discussed, but no data were presented; namely, the "letter grade" (i.e., A, B-, C+, F, etc.) used by many schools. This idea was appealing for two reasons. One, the audience who generally needed an absolute value associated with a SUS score (e.g., project managers, designers, etc) would almost universally recognize it. Also, as shown in Bangor et al. (2009), standard scoring (e.g., 90+ for an A, 80-89 for a B, etc.) matched up reasonably well with other ways to determine acceptability of scores. However, Sauro and Lewis (2012) pointed out that using standard grade scoring seemed unusually strict, given the distribution of SUS scores across the 0-100 range. They proposed grading on a "curve" and proposed SUS score ranges that might better fit the percentile data. Still, this was an after-the-fact fitting to the SUS data and even if it may have been a useful "rule of thumb," how actual users would grade a product and how that result would match up to the SUS score provided remained unknown.

Method

As was done with the adjective-anchored rating scale (Bangor et al., 2009), for the letter grade assignment an eleventh statement was added at the end of the modified SUS survey. The following statement was presented to participants who completed the SUS: "Please assign a letter grade to the user-friendliness of this product."

The letter grade statement was used in 15 usability studies, consisting of four to six tasks with an Interactive Voice Response (IVR) system used by callers for customer service related tasks (e.g., billing and payments, account issues, technical support, etc.). Numeric scoring for letter grades consisted of 4.0 for an A or A+, 3.7 for A-, 3.3 for B+, 3 for B, 2.7 for B-, and so on down to 0 for an F.

Results

The modified SUS including the letter grade statement was completed by 187 participants (91 Females; Age: M=43.3, SD=14.6). SUS scores and the numeric equivalent for letter grades were highly correlated (r=0.848). Figure 1 shows the compiled results, i.e., the mean scores (error bars are +/- one standard error of the mean).

Copyright 2013 by Human Factors and Ergonomics Society, Inc. All rights reserved. DOI 10.1177/154193121357104

Downloaded from pro.sagepub.com at HFES-Human Factors and Ergonomics Society on June 3, 2014

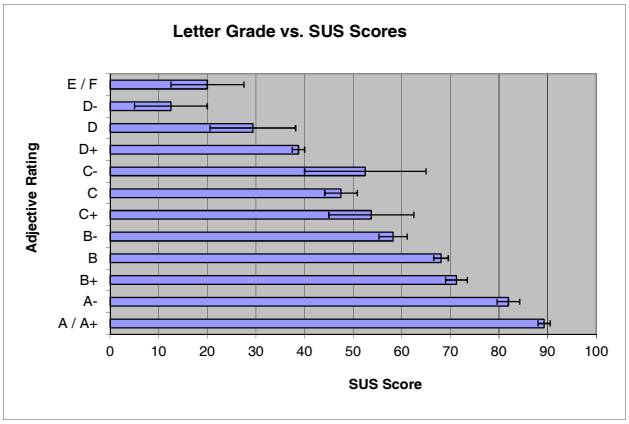


Figure 1. Depiction of mean SUS scores based on letter grades assigned.

Discussion

The data in Figure 1 seem to indicate that participants expanded their letter grade assignments to fit nearly the full range of possible SUS scores. This is similar to the findings with the adjective data from Bangor, et al. (2009). The data suggest participants are much more generous with the letter grades than the proposed ranges given by Bangor, et al. (2009) and Sauro and Lewis (2012). It is unclear whether participants in the studies were giving absolute ratings or were just providing another rating along an end-to-end scale. What is clear is that higher SUS scores are linked with products that are more user friendly. More research is needed into developing benchmarks.

USING THE SUS WITHOUT DIRECT TASKS

One of the classic uses of the System Usability Scale is in a test-then- measure scenario. The experimenter develops a task, or a set of tasks for a user to perform with a specific product or service. After completion of those tasks, the user then rates the usability of the product using the SUS. If the experimenter aims to understand specific sub-actions within a product, the task may be narrowly defined for the user (e.g. "check to see if you have email in Outlook"). If the practitioner is interested in a more global rating of the product, the task set may be constructed quite broadly to more fully exercise the features and interface of that product. In either case, the user has a specific reference set of tasks upon which to base his or her rating of the product with the SUS. Further, because all users in a typical usability test will perform the same set of tasks, the experimenter has a common frame of reference upon which to base the interpretation of the resulting SUS data.

Most of the published work that has utilized the SUS has used this test-then-measure paradigm. (Költringer & Grechenig, 2004; Greene, Byrne, & Everett, 2006; Bangor et al., 2008; McLellan, Muddimer, & Peres, 2012). This is the gold-standard use for the SUS because it allows for the direct comparison of competing products when users perform similar or identical tasks, it provides a high degree of experimental control and it can generate effectiveness and efficiency metrics as recommended in ISO 9241-11 (ISO, 1998). However, we suggest that it is not always necessary to use this test-then-measure paradigm for the evaluation of product and services. In fact, there are several situations where using the SUS in a less traditional manner can be beneficial.

Retrospective evaluation of products and services

It is often true that users of particular products and services use those products in very different ways. If a product is complex or feature-rich, a user may only have the need for a limited subset of features, or may not have the expertise to utilize specific features of the product. For example, while MS Excel's fundamental spreadsheet functions are relatively simple to use, some of the more advanced features, such as Pivot Tables and the Solver functions may be beyond the skills or needs of many users. If a usability professional were to attempt to explore the overall usability of MS Excel, one of the first steps would be to devise a core set of features on which to base the subsequent usability measures. What would comprise this core functionality? Because of the feature-rich nature of the product, a representative task set that exercises the whole product would be time-prohibitive. Experienced users in each of the specific sub-tasks could also be brought in to test individual components, but that minimizes the learning component of the SUS.

Customizable interfaces are another example where standard test-then-measure protocols may not produce the needed data. Having users test other's special customizations puts them at a significant disadvantage and testing a series of users on their own customized interfaces eliminates the advantage of common tasks in the subsequent analysis, making metrics such as efficiency and satisfaction difficult to interpret.

If instead we simply ask users to assess the usability of a specific product, without a preceding task, then users will be free to base their assessment on an integrated, retrospective view that reflects their entire prior experience with the product. Expert users will base their assessment of the product on the sum total of all the features that they employ, and novice users will do the same. Factors such as the ease of learning the interface over time can be considered and weighed against the ease of use for commonly performed functions by each user for his or her specific experiences. When using this paradigm, it is important to guard against recency effects (which of course can exist even in well controlled test-then measure experiments) by reminding the user to consider all of their experiences with the specific product when making their SUS ratings. Of course, this paradigm has the added benefit of being extremely efficient to use, as multiple products can be assessed by users without the need for direct supervision or the time necessary for users to perform series of tasks. This allows for large numbers of users to participate in the study in a cost effective fashion.

Evaluating classes of products

Sometimes the question is not how usable a *specific* product is, but rather how usable a specific *class* of interfaces is. For example we might want to distinguish between broad segments of products within a class (e.g. are laser, inkjet or all-in-one-printers easier to use?) in order to help guide product development efforts. We might also want to collect this kind of data to help guide our selection amongst a number of specific interfaces. For example, if we are going to field a limited number of customer facing interfaces due to cost or time constraints, we might want to develop and deploy those

Downloaded from pro.sagepub.com at HFES-Human Factors and Ergonomics Society on June 3, 2014

that are easiest for our specific customer base to use (e.g. should I deploy an IVR or smartphone interface?)

Another use of product class ratings is to help frame the usability of a specific product against a class of well-known products. For example, a number of cell phone distraction studies frame the performance of participants by comparing their performance levels to those established in performance studies of drivers who have elevated blood alcohol levels (e.g. Strayer, Drews & Crouch, 2006). This kind of assessment allows consumers to use their previous experiences with well know interfaces or situations to make meaningful comparisons to an unknown product or service.

Measuring classes of product is no different than measuring a specific product – the user is simply asked to assess the usability of a class of product, like microwaves, rather than a specific model, based on the totality of their experiences with that product class. In many instances, this is a benefit in measurement, since users often do not have an idea of which specific brand or model of product they use, but can render an opinion on the usability the device in general. As with the no-test protocol described above, the method has the advantage of integrating a user's experience across all of their interactions with the product, in proportions that are weighted to their own use patterns.

When employing the SUS in this fashion, it is important to keep in mind the limitations of the data that are being collected. You need to be aware of, and able to live with, the uncertainty of the user's task definition. Maybe some of the complex, hard to use features are never considered or evaluated by users because they never discover or use them. Does that make the evaluation of the product or class of products by the user any less valid? We would argue that it does not, because the user is evaluating the product for the tasks *they* perform in the time and manner in which they themselves choose to perform them.

Both of these approaches have been used in the field with good success (Kortum & Bangor, 2013) and in ongoing research into the usability of mobile applications. If the goal is to establish benchmarks or to capture the sum total of a user's experience with a specific product or even a class of products, then the task-less measurement of usability may have some utility for practitioners and researchers alike.

ASSESSING POTENTIAL ERGONOMIC IMPACT: WHERE THE SUS SHOULD NOT GO

Over the last two decades, many ergonomists have found, and computer users have experienced, that extended computer use can cause discomfort and even injury in the upper extremities. Indeed in 2007, Taylor found that up to 40% of reported lost-time incidents may have been related to computer usage (Taylor, 2007). In 2010, repetitive motion injuries was identified as one of the top ten most disability injuries—causing 4.0% of the injuries that required an employee to miss six or more days from work and costing $2.02 billion in workers compensation insurance (Liberty Mutual Research Institute for Safety, 2012). Previous efforts to minimize these injuries have included environment adjustments (e.g., adjusting people's chairs, keyboards and monitors to allow for them to be in neutral posture while using the computer) and behavioral change (e.g., taking frequent breaks, stretching, etc.). It appears that many of these efforts have been successful because LMRIS also reported that the number of repetitive strain injuries (RSI) has decreased by 39.7% from 1998 to 2010 (Liberty Mutual Research Institute for Safety, 2012). However, given that this type of injury still accounts for 4% of injuries, there is clearly more work to do to identify and mitigate the causal factors associated with them.

A multidisciplinary group began to investigate whether the design of the software people were using could in fact be contributing to the rate of these injuries. To do this, we began a series of studies focused on evaluating the elements of software and whether they differentially impacted RSI risk factors. However we quickly realized that typical measures of evaluating software (i.e., the SUS) had been developed as a measure of how easy it was for a person to accomplish a task but not necessarily as a measure of the discomfort or the muscle activity that occurred while using the software. To measure muscle activity, researchers typically use surface electromyography (sEMG), which involves placing electrodes on the surface of the skin just above the muscle to assess muscle activity (Freivalds, 2004). Although a valid and reliable measure of muscle activity while someone is performing a task, this method is expensive and time consuming.

To explore the relation between software design and risk exposure to RSI in real world settings (e.g., in the iterative design lifecycle with developers), we needed a low cost and easy to use measure of muscle activity. This typically translates to a self-report measure. However, the measure also needed to be validated and reliable. We identified four self-report measures that might work. Each of these measures has been use to evaluate physical activity in several different settings and each results in a single, one-dimensional value.

- NASA Task Load Index (TLX: Hart & Staveland, 1988)
- Latko Busiest Hand Activity Scale (Latko: Latko, Armstrong, Foulke, Herrin, Rabourn, & Ulin, 1997)
- BORG Scale of Perceived Exertion (Borg: Chan, Chow, Lee, To, Tsang, Yeung, & Yeung, 2000)
- System Usability Scale (SUS: Sauro, & Lewis, 2012).

However, none of these measures has been validated against sEMG as a measure of muscle activity or strain particularly for office ergonomic applications. To begin the process of validating these measures for muscle activity, we conducted studies in a laboratory environment at the University of Houston-Clear Lake (UHCL) as well as in the field with geoscientist at petroleum companies in Houston, Texas. As reported in Peres, Kortum, Muddimer and Akladios (under review), we found small but reliable correlations between muscle activity and the Latko, Borg and TLX. Our design did not allow us to calculate correlations between the SUS and muscle activity directly, so we instead re-examined the data from the UHCL laboratory studies and the field study to explore whether the SUS correlated with the Latko, Borg and TLX. Additionally, a laboratory study was conducted at Rice to explore the correlations between the self-report surveys and the SUS with different applications than those in the previous studies.

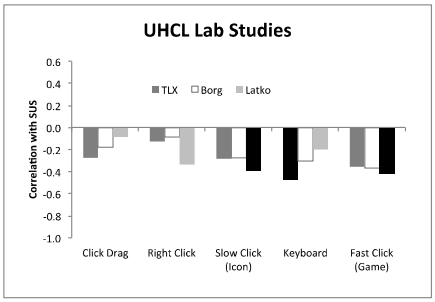
Downloaded from pro.sagepub.com at HFES-Human Factors and Ergonomics Society on June 3, 2014

UHCL

Method: Twenty-seven UHCL students (22 females) were recruited from the UHCL participant pool to participate in the experiment (Age, *M*=27.6, *SD*=9.5). They sat at a computer workstation and were instructed to adjust the workstation so it was comfortable. Participants then edited a MSWord document using whatever method of interacting with the software that they wanted. They then completed five counterbalanced tasks using an interaction method specified by the experimenter. Four of the five tasks involved participants performing editing tasks in MS Word such as *find* and then *bold* a word using the keyboard, icon (Slow Click), mouse right-click (Right Click), or mouse dragging (Click Drag). Another section involved the participants playing a web-based game called "Hit-the-Dot" which requires people to use the mouse to successively and quickly make mouse clicks (Fast Click). After each task the participants complete the Latko, Borg, and TLX. At the end of the study, participants were asked to evaluate MSWord using the SUS.

Results: Figure 2 shows the correlations between the SUS and the three additional self-report surveys. The three black columns represent correlations that were significant at the 0.05 level.

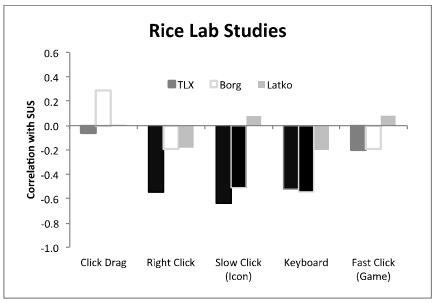
The significant correlations were for the TLX for the Keyboard task and the Latko for the Slow and Fast Click tasks. Further, all of the correlations are negative which is the expected direction, i.e., as usability (SUS) increases, discomfort decreases.



Rice University

Method: Thirty Rice University students (8 females) were recruited using the Rice participant pool to participate in the experiment (Age: *M*=19.8,*SD*=1.05). Participants sat at the computer workstation and were given instructions for five counterbalanced tasks using five different applications. The five applications required participants to perform tasks that used the same interaction techniques as the tasks used in the study in the UHCL lab studies. Upon completion of each task, participants completed the SUS, Latko, Borg, and TLX.

Results: Figure 3 shows the correlations between the SUS and the three self-report measures. The black bars represent those correlations that are significant at the 0.05 level. The correlations between the SUS and the TLX were significant for the Right Click, Slow Click and Keyboard tasks. Between the SUS and the Borg, there were significant correlations for the Slow Click and Keyboard tasks. There were no significant correlations between the SUS and the Latko.



Field study

Method: Twenty-four geoscientists spent at least an hour conducting their work. After this period of time, the participants completed the SUS, Latko, Borg, and TLX and were asked to complete these based on the primary work they had been doing during the last hour.

Results: Table 1 shows the correlations between the SUS scores and the three self-report surveys. None of the correlations were significant and were all virtually zero.

Table 1. Correlations of three self-report surveys with the SUS (all p's >0.70)

TLX	0.010
Borg	-0.062
Latko	-0.030

Summary

The results from these three studies strongly suggest that the SUS is not a reliable measure of the discomfort or muscle activity associated with interacting with software. Although both of the laboratory studies found some significant relations between the SUS and the self-report measures, the only consistent finding between the two was the correlation between the TLX and the SUS for the keyboard-based interactions. Additionally, the fact that the correlations between the surveys and the SUS were essentially zero for the Field study, provides more support for the notion that the SUS is not a valid tool for assessing discomfort.

There are several important points to consider however, when evaluating these results: 1) the studies have several fundamental differences in their designs. For instance, in the Rice study, participants completed the SUS for every task, while they only completed it for a period of time in the other studies. 2) The TLX, Borg, and Latko are reliably associated with muscle activity, but as stated earlier, this relationship is not strong (correlations of approximately .15). Therefore it is possible that if the SUS were correlated against a measure that was more strongly related to muscle activity that the result would be different. And 3) the SUS scores for all three studies were relatively high (UHCL: M = 79.4; Rice: M = 76.0, and Field: M = 77.4). This restriction of range could have reduced the likelihood of our finding a relationship between the SUS and the self-reports that was actually there.

Despite the potential weaknesses of these studies, these findings clearly illustrate that currently the SUS is not an appropriate measure for investigating the potential ergonomic impacts of software interaction techniques.

Downloaded from pro.sagepub.com at HFES-Human Factors and Ergonomics Society on June 3, 2014

CONCLUSIONS

In summary, the SUS is a valuable tool for assessing commercial usability either in a standard usability test or when users are asked to recall their experiences with a product or class of products. Usability however, does not include ergonomics. Though, more research is needed to define benchmarks for usability. The remaining papers in this symposium continue the discussion of valid uses of the SUS. Specifically, Peres reports on the relationship between SUS scores and performance data. Kortum describes the impact of user experience on SUS data. Bangor discusses how SUS data can be tied to metrics identified as key to business success for wireless handsets. Finally, Grier examines the potential use of the SUS in military acquisition.

REFERENCES

Bangor, A., Kortum, P. & Miller, J.A. (2008). The System Usability Scale (SUS): An empirical evaluation. *International Journal of Human-Computer Interaction*, 24(6),574-594.

Bangor, A., Kortum, P. & Miller, J.A. (2009). Determining What Individual SUS Scores Mean: Adding an Adjective Rating Scale. *Journal of Usability Studies*, 4(3), 114-123.

Borsci, S., Federici, S., & Lauriola, M. (2009). On the dimensionality of the System Usability Scale: a test of alternative measurement models. *Cognitive Processes*, 10, 193-197.

Brooke, J. (1996). SUS: A 'quick and dirty' usability scale. In Jordan, P.W., Thomas, B., Weerdmeester, A. & McClelland, I.I. (Eds.) *Usability evaluation in industry, london* (pp 189-194). Taylor & Francis.

Chan, R., Chow, C., Lee, G., To, L., Tsang, X., Yeung, E., & Yeung, S. (2000). Self-perceived exertion level and objective evaluation of neuromuscular fatigue in a training session of orchestral violin players. *Applied Ergonomics,* 31, 335-341

Freivalds, A. (2004). *Biomechanics of the upper limbs: Mechanics, modeling and musculoskeletal injuries.* Boca Raton, FL: CRC Press.

Greene, K.K., Byrne, M.D. & Everett, S.P. (2006). A comparison of usability between voting methods. *Proceedings of the USENIX/Accurate Electronic Voting Technology Workshop 2006 on Electronic Voting Technology Workshop EVT'06* (p. 2). Berkeley, CA: USENIX Association.

Hart, S. G. & Staveland, L. E. (1988) Development of NASA-TLX (Task Load Index): results of empirical and theoretical research. In P. Hancock & N. Meschkati (eds.). *Human mental workload* (pp 139-183). Amsterdam: Elsevier Science.

ISO. (1998). *Ergonomic Requirements for office work with visual display terminal (VDT's) – Part 11: Guidance on Usability* (ISO 9241-11(E)). Geneva, Switzerland.

Költringer, T. & Grechenig, T. (2004). Comparing the immediate usability of graffiti 2 and virtual keyboard. In *CHI'04 extended abstracts on human factors in computing systems* (CHI EA '04; pp 1175-1178). ACM, New York, NY, USA.

Kortum, P. & Bangor, A. (2013). Usability Ratings for Everyday Products Measured With the System Usability Scale (SUS). *International Journal of Human Computer Interaction,* 29, 1-10.

Kortum, P., Grier, R.A., & Sullivan, M. (2009). DSL Self-installation: From Impossibility to Ubiquity. *Interfaces,* 80, 12-14.

Latko, W., Armstrong, T., Foulke, J., Herrin, G., Rabourn, R. & Ulin, S. (1997) Development and evaluation of an observational method for assessing repetition in hand tasks. *American Industrial Hygiene Association Journal,* 58, 278-285

Liberty Mutual Research Institute (2012) *From Research to Reality, Scientific Update, Winter 2012.*

McLellan, Muddimer & Peres (2012). The effect of experience on System Usability Scale Ratings. *Journal of Usability Studies*, 7(2) 56-67.

Peres, S. C., Kortum, P. K., Muddimer, A., & Akladios, M. (Under Review) Developing and Validating a Self-Report Assessment Tool for Software Ergonomics. *Applied Ergonomics.*

Sauro, J., & Lewis, J. R. (2012). *Quantifying the User Experience: Practical Statistics for User Research.* Burlington, MA: Morgan Kaufmann.

Strayer, D.L., Drews, F.A. & Crouch, D.J. (2006). A Comparison of the Cell Phone Driver and the Drunk Driver. *Human Factors,* 48(2), 381-391.

Taylor, K. (2007). Oil industry dangers lurk in the office environment: *Oil & Gas Network Magazine,* April 2007.

The Panelists

Dr. Rebecca Grier is an applied experimental psychologist with over 10 years experience working to improve Human Systems Integration (HSI) in military & commercial systems. Currently she is a Research Staff Member at the Institute for Defense Analyses (IDA). In this position, she conducts social science research on issues of importance to the U.S. government and supports the test and evaluation of DoD systems. Previously, at Naval Sea Systems Command, Dr. Grier was the acting Technical Warrant Holder for Displays and Controls and the HSI lead for Aegis Modernization. She has also worked at Aptima, Inc. & SBC Technology Resources Inc. (now AT&T Laboratories). Dr. Grier holds a Ph.D. & M.A. in Human Factors/ Experimental Psychology from the University of Cincinnati & a B.S. Honors in Psychology from Loyola University, Chicago.

Dr. Aaron Bangor is the Lead Accessible Technology Architect at AT&T. There he conducts research & promotes ways to make technology usable by all. Previously, he worked in the Human Factors Engineering Group of AT&T Labs where he conducted research to find out how to make technology more user-friendly. He earned a Ph.D. in Human Factors Engineering from Virginia Tech, currently holds 11 patents, & is a Certified Human Factors Professional. Dr. Bangor also serves as the Vice-Chair of the Texas Governor's Committee on People with Disabilities. He is active in the area of standards, including part editor of ANSI/HFES 200 & is a member of the U.S. Technical Advisory Group to the International Standards Organization's TC 159 (Ergonomics) SCs 1 & 4.

Dr. Philip Kortum is currently a faculty member in the Department of Psychology at Rice University in Houston, Texas. Prior to joining Rice, he worked for almost a decade at SBC Laboratories (now AT&T Laboratories) doing human factors research & development in all areas of telecommunications. Dr. Kortum continues to do work in the research and development of user-centric systems in both the visual (web design, equipment design, image compression) & auditory domains (telephony operations and interactive voice response systems). He received his Ph.D. from the University of Texas at Austin**.**

Dr. S. Camille Peres is an Assistant Professor at the Texas A&M University. Her research is focused on the design implications for Auditory Displays, the ergonomic implications of input methods (gestures, touch, etc), & incorporating simulations into the teaching of statistics. Dr. Peres received a Masters from the University of Houston-Clear Lake in 1996 and a Masters and Ph.D. in Psychology from Rice University in 2005. She is the incoming president for the Houston Chapter of HFES & is the Division Chair for the Professionalism Division of HFES.

Downloaded from pro.sagepub.com at HFES-Human Factors and Ergonomics Society on June 3, 2014